

⚡ 08. The Problem with Async Errors in Express

💣 Why Async Errors Break Express 4

Synchronous code me agar error throw ho toh Express ka default handler usse catch kar leta hai. Lekin **async functions me unhandled rejected promises Express catch nahi kar pata!** Request hang ho jati hai ya process crash ho jata hai.

😓 The Ugly Solution: Repetitive Try-Catch

Har ek async route handler me try-catch lagana code ko bohot cluttered aur unmaintainable bana deta hai:

Ugly Try/Catch Pattern

```
app.get("/chats/:id", async (req, res, next) => {
  try {
    let chat = await Chat.findById(req.params.id);
    if (!chat) throw new ExpressError(404, "Chat Not Found!");
    res.render("show.ejs", { chat });
  } catch (err) {
    next(err); // Har route me catch likhna padta hai!
  }
});
```



BOILERPLATE OVERLOAD: Real projects me 50+ routes hote hain. Har jagah try/catch likhne se code readability khatam ho jaati hai!

🌟 The Pro Solution: wrapAsync & Mongoose Errors

🌟 The wrapAsync Higher-Order Function

wrapAsync ek higher-order function hai jo async function ko wrap karke kisi bhi error ko automatically `.catch(next)` me forward kar deta hai!

```
utils/wrapAsync.js - Magic Wrapper

function wrapAsync(fn) {
  return function (req, res, next) {
    fn(req, res, next).catch(next); // 🐞 Automatically passes rejected promise to next(err)
  };
}
module.exports = wrapAsync;
```

🌟 Clean Route Implementation

```
index.js - Clean Async Route

// No try-catch needed! Pure clean, readable code:
app.get("/chats/:id", wrapAsync(async (req, res) => {
  let chat = await Chat.findById(req.params.id);
  if (!chat) throw new ExpressError(404, "Chat Not Found!");
  res.render("show.ejs", { chat });
})));
```

🎯 Handling Specific Mongoose Errors

Mongoose Error	Trigger Reason	Handling Technique
ValidationError	Schema validation fail (required field missing)	Extract <code>err.errors</code> and send 400 Bad Request.
CastError	Invalid ObjectId passed in URL (e.g. /chats/123)	Check <code>err.name === "CastError"</code> and return 400 "Invalid ID".
MongoServerError (11000)	Duplicate key error (e.g. email already exists)	Check code 11000 and return "Email already registered".